Developing ICU-Talk - A Computer based Communication Aid for Patients in Intensive Care

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Abstract

This paper highlights the development of a computerised communication aid for intubated patients in an Intensive Care Unit (ICU).

Intubated patients in ICU have few methods of communicating other than attempting to mouth words and use gestures. Communication aids that are available, such as symbol charts and alphabet boards, unfortunately have not been designed to meet the specific requirements of these patients. Even these methods are not ideal for the patient, their family members or the staff caring for the patient. From previous research studies of communication within ICU, patients often feel disempowered and depressed due their inability to contribute to conversations. This often leads to feelings of social isolation. ICU-Talk will be developed to allow the patient to have more control of their communications and at the same time address their ongoing problems and needs on an individual basis.

Introduction

The ICU-Talk project is a three-year collaborative project which aims to develop and test a computer based communication aid designed to meet the specific needs of intubated patients in Intensive Care Units. The project has been funded by the Engineering and Physical Science Research Council (EPSRC). The development team for the ICU-Talk system consists of a senior staff nurse from ICU, a speech and language therapist, a software engineer and a doctoral student in computing. The multi-disciplinary nature of this team means that a range of skills and knowledge including clinical experience of patient care, knowledge of human communication patterns, assistive communication devices, and software/interface design have been brought together to ensure all these aspects are considered in the design, implementation and evaluation of ICU-Talk.

Background

The use of communication aids with patients who are intubated and/or ventilated as in ICU has always been restricted. These patients are unable to talk because of a tube in their nose, mouth or throat (endotracheal tube) which passes into their windpipe (trachea) and helps with breathing but temporarily stops the vocal cords from working. They are also often very unwell, with acute illnesses or severe injury following an accident. Many patients admitted to an ICU only stay for a short time and may be sedated throughout. However, there is a group of patients whose stay in ICU is prolonged. These patients may be awake and orientated but unable to communicate effectively. Weakness in their hands and arms means they are unable to write or use gestures. This group of patients present a particular set of problems as they attempt to communicate with health care personnel, friends and relatives. Their attempts at mouthing words or using facial expression are often misinterpreted or are simply not understood. This situation can lead to feelings of frustration, isolation, stress and inadequacy and can affect the patient's recovery [2], [3].

Currently available Augmentative and Alternative Communication (AAC) devices have been designed either for clients with language impairment and/or learning difficulties or for clients with intact language but a physical disability. No commercially available AAC devices have been designed with the specific needs of the ICU patient in mind. The patient's medical condition and the high levels of stress and anxiety experienced by patients in ICUs mean they tire very quickly. These factors also affect their cognitive skills including their ability to learn, retain information and concentrate on tasks. This in turn affects the patient's ability to use an AAC device. Many patients may also have physical deficits. These can include weakness, loss of sensation and reduced movement of the hands, arms, legs, feet and head. This makes accessing any AAC device problematic, as the patient may not be able to use a keyboard, mouse, trackball or touch screen. Finding a reliable movement for use with a single switch may be possible but currently available switch operated AAC systems are either highly cognitively loaded or designed only to convey very basic needs. Given these challenges it is not surprising that the use of high-tech computer based communication aids in ICU has been generally unsuccessful. Low-tech communication aids e.g. alphabet charts, picture boards and eye pointing systems are used along with mouthing, writing, gesture and facial expression, but these systems are time consuming, frustrating and inaccurate [1].

Nursing staff in ICU are highly skilled in facilitating communication with intubated patients. They use combinations of yes/no questions to try and determine patients' needs and wants and with experience they become proficient at anticipating and interpreting what the patient is attempting to communicate. This works reasonably well with needs based communication but becomes much more difficult when the nurse and patient do not share common knowledge about a subject or if the topic of conversation is unrelated to the patient's immediate environment. For patients who have a prolonged stay in ICU this reduction in communication will affect their ability to build and maintain relationships and social closeness with friends, family, nurses, and other members of the hospital team.

The TalksBac system

An augmentative communication system, designed specifically for and tested by adults with dysphasia, was developed during four years of collaborative research involving Dundee Speech & Language Therapy Service and Dundee University's Applied Computing Department [5]. This system is called TalksBac. TalksBac allows the user to access a database of pre-stored sentences and stories. Its interface is designed to be simple to operate and navigation through the system can be achieved by mouse, trackball or touch screen. TalksBac also has the potential to support a simple scanning interface (see Figure 1). TalksBac uses prediction to help guide the user through their interaction. The transparency of the TalksBac system makes it useful for patients who have a short concentration span and whose ability to learn and remember is compromised. It was felt that the design principles used for TalksBac could also be applied to help patients in ICU communicate more effectively. To asses this a pilot TalksBac database was developed which illustrated the ways in which such a system could support a patient in ICU. Feedback from the ICU consultant and nursing staff provided positive support for developing the ICU-Talk system.

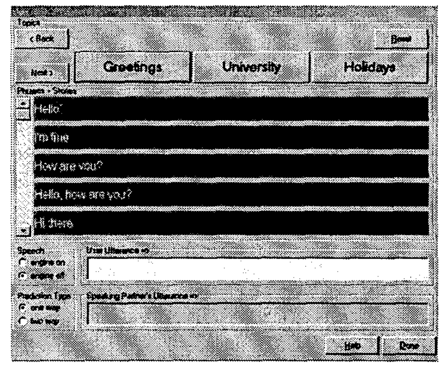

Figure 1. Screenshot of TalksBac interface.

Creating an initial database of patient utterances

Nursing questionnaire

The ICU nursing staff were interviewed to obtain an initial list of phrases normally used by ICU patients. Using existing clinical expertise the development team was able to identify nine commonly used topics of conversation:

- Environment
- Needs
- Feelings
- Family and friends
- Work
- Medical
- Future-short term
- Future-long term
- Other

A short structured interview was assembled and thirty-two of the forty-two ICU nursing staff were interviewed over a period of three weeks. Each nurse was asked to provide examples of three or more, questions or comments that patients usually say within a given topic.

For example:

Environment Topic- *"It is too hot" or "Where am I?"*
Needs Topic- *"I need a drink" or "I need a cough"*
Medical Topic - *"When will I get my tube out?"*

The development team combined similar phrases, questions and comments. The responses from the nursing staff formed a core database of one hundred and ninety phrases arranged under the nine topics listed above. Some topics contain more phrases than others. A data recorder containing these results was developed for the purpose of observing patients attempting to communicate in ICU.

Patient selection

Having created the core database, the next phase was to observe suitable intubated patients in ICU who were trying to communicate and hereby establish what patients were actually trying to say (see figure 2).

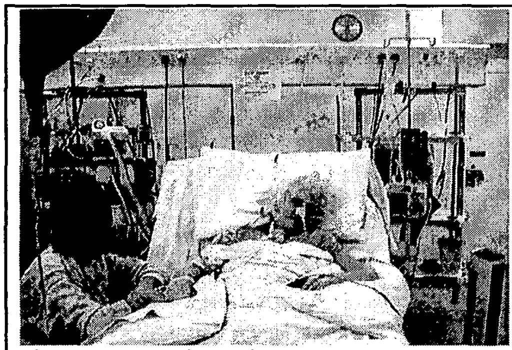

Figure 2. Patient in ICU attempting to communicate with daughter.

An "ideal" patient profile, in the form of a flow chart, was compiled to help nursing staff identify suitable patients. The flow chart prompts the nursing staff with a series of questions. If at any point the answer to a question is "no", then the patient is not suitable for inclusion in the project. The questions asked were:

- Is the patient intubated?
- Is the sedation score between 1 and 3?
- Is the patient able to consistently respond to a basic command?
- Is the patient attempting to communicate?
- Is the patient able to focus?
- Was the patient literate prior to admission?

Nursing staff will be able to use the flow chart, after the project has ended, to identify patients who may be able to use a communication aid.

The sedation score is a scoring system that is unique to Ninewells ICU. An assessment of the patient's level of consciousness is made and a score of between 1 and 6 given to that patient [4]. Patients who are able to communicate have a sedation score of between 1 and 3.

Sedation scoring system:

1. Alert and anxious.
2. Aware and calm.
3. Drowsy and co-operative.
4. Asleep; responds to verbal command or light glabellar tap.
5. Asleep; responds to painful stimuli.
6. Unconscious and unrousable.

After referral by the nursing staff, the project development team assessed the patient and decided if the patient was suitable for inclusion within the project. The patient's permission to participate with the project was also obtained. On acceptance, the project aims were explained to the patient and they received a demonstration of the data recorder system. Support from the patient's family or friends was sought and they were involved with discussions about the project and the expected outcomes. Only phrases communicated by the patient were recorded and the nursing staff, family and friends were made aware of this.

Patient observation

A computerised core data recorder was developed to assist with observation of communication by selected patients in ICU. The recorder holds the core database of phrases provided by the nursing staff and the observer matches the patient's communication attempts to these or adds any novel phrases.

The observer records "who" the patient is trying to communicate with e.g. family, nurse, doctor, and "how" the patient communicates e.g. mouthing, gesturing, or alphabet board. Then for each patient communication, the observer selects the topic and the phrase communicated. Each phrase used by a patient is recorded in the observer's personal event trace file. The trace includes who the patient spoke to, how they did it, what they said and when it was said. If the phrase that the patient communicates is not contained within the database then the observer will add it under the appropriate topic.

Early analysis indicates that patients say very individual things but there are common phrases, such as " I need a cough", " I need a drink" and "Thank you".

Using this process a increasing representative core database will be compiled containing the most commonly used patient phrases.

Results

The results detailed below are taken from 12 ICU patients. These patients were observed for a total of 30 hours. The results were compiled from the event trace file and the database files that the observation software uses and maintains.

Phrases

A total of 406 phrases were observed and recorded. 178 of these were phrases were present in the core database. This represents 25% of the core database. The remaining 228 phrases were patient specific. Figure 3 shows the 21 most frequently communicated phrases.

Who

The person with whom the patient was communicating was recorded for each phrase. The most common communication partner was the nurse. The patient communicated with the nurse 44% of the time. The next most common communication partner was a family member. Although there are other staff in ICU they

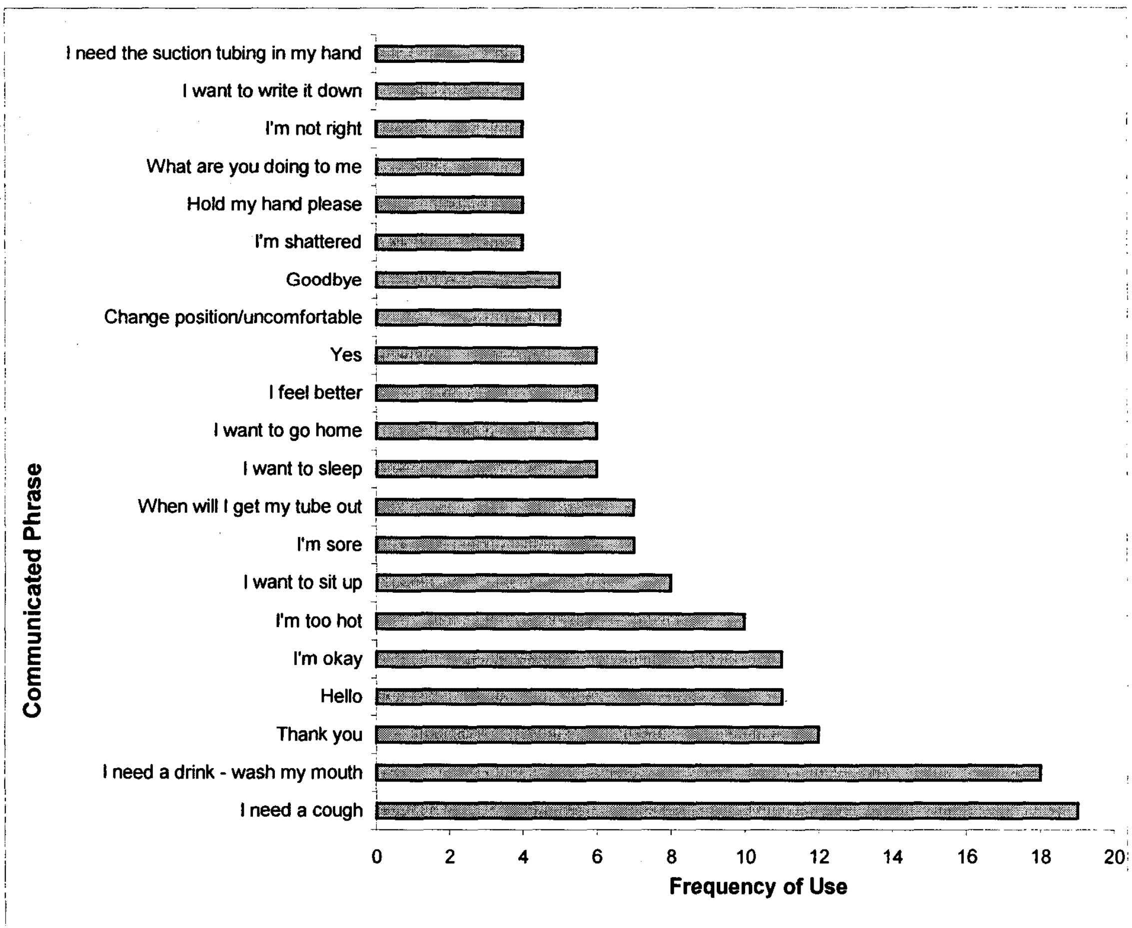

Figure 3. Phrases most frequently communicated by patients.

normally do not talk directly to the patient but instead communicate via the patient's nurse or relative. Their communications are such that they rarely give the patient the opportunity to ask questions or engage in conversation.

Topics

The phrases were organised under nine topics. Feelings and Needs were the topics most frequently used. From the observation results, the topics "future-long term" and "future-short term" were combined to form the new topic "future". The number of possible conversational topics as indicated in the figures below, now contains eight topics.

How

The method used by the patient to communicate a phrase is referred to as "how". How the patient communicated each phrase was also observed and recorded. If more than one method was used to communicate then the predominant method was recorded. Mouthing was used most often by patients (63%) followed by gesture (25%). The term gesture was used to describe a physical gesture or facial expression used in a communicative way. These results reflect the patient's natural reaction to try to use speech to communicate even though they know they are unable to achieve a voice. Many patients had severe weakness of their upper limbs and were unable to write, produce specific gestures or point accurately to an alphabet chart.

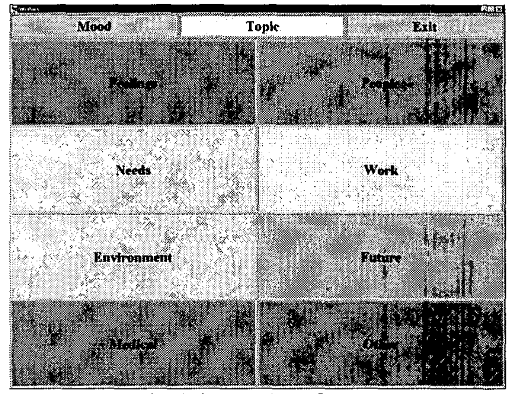

Figure 4. Standard (boxes) interface.

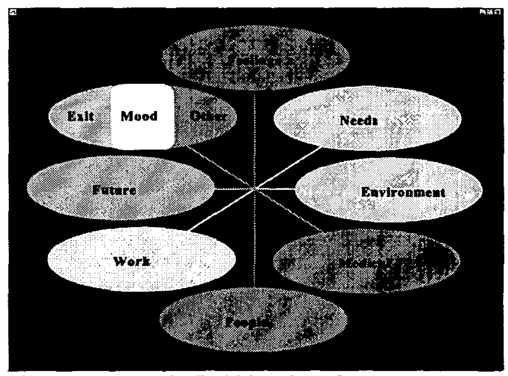

Figure 5. Dynamic (bubbles) interface.

Interface design

The design of a suitable interface is central to the ICU-Talk system. Navigation by a patient to select an utterance needs to be fast, easy to learn and memorable. Advice on interface design was sought from a local software development company and principles from computer game design incorporated into ICU-Talk to make it both visually stimulating and intuitive in use. One of the important underlying principles was consistency, e.g. If a section of the program was to present displays as sliding from left to right then this had to be continued throughout the program. Similarly orders of buttons and arrows should also illustrate the movement of left to right The specific needs of the target patient group also need to be considered at each stage of the interface development. It was decided that the patient should be able to choose which interface they prefer from a choice of two styles, standard (boxes) or dynamic (bubbles), (see figures 4 and 5). Both interfaces link to the same underlying algorithms and data structures. Patients who will benefit from ICU-Talk are likely to have a combination of physical and cognitive problems [1] and this has influenced the interface design.

The ICU-Talk system has been designed so that minimal training is required and phrases are ordered to allow access with the minimum number of selections. As part of the training the patient will be asked to set some of the interface parameters. These parameters will include the sensitivity of the touch screen, speed of the mouse emulator and the background colour, font size and style for the phrase buttons. This means that the patient can customise the interface to a limited degree and so have some control over the appearance of the system.

When starting each session with ICU-Talk the patient is asked to select how they are feeling from a choice of three icons that represent their mood. They then select an energy level from a choice of six. The energy level represents how tired or how full of energy the patient feels (see figure 6). It is used together with the patient's mood selection to allow the system to order phrases e.g. if the patient's mood

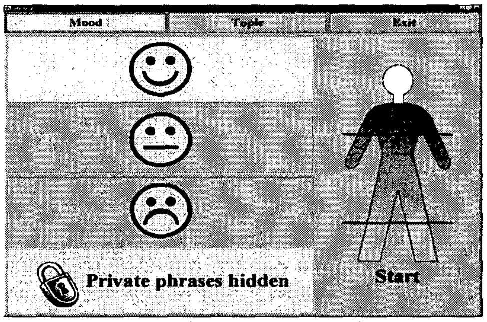

Figure 6. Mood and energy screen selector.

is happy and combined with high energy levels then positive phrases will be placed at the top of the phrase lists.

The next selection the patient makes is the topic of conversation. The interface containing the topic choices has easy to read buttons that provide a large target area for selection by touch screen or mouse emulator. The system also has the facility for any single-switch user to use a scanning system to navigate the interface.

The topic buttons are colour coded so the patient can build up an association between the colour and the topic to aid their navigation through the system. Once the topic is chosen a list of potential phrases is displayed (see figure 7). The background colour of the phrases screen is the

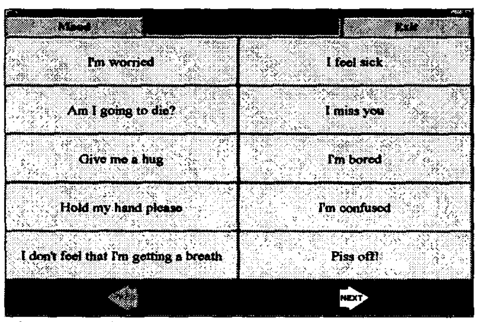

Figure 7. Phrase selector screen.

same as the topic colour. The colour of the phrase buttons, the size and style of font of the phrases is determined by the patient.

The ICU-Talk system also includes the facility to have 'private' phrases hidden from general view. The patient can select the icon (a picture of a padlock) on the interface, which will then allow these additional phrases to be displayed on the screen. In many commercially available AAC devices all phrases are displayed on the screen and are potentially available for everyone to see. By keeping private phrases hidden from view the patient can have more control over conversations, and build and maintain relationships in a more normal way e.g. the patient may have things they want to say to their spouse but not to the staff in ICU.

Patient evaluation of the system

The iterative nature of the design of the ICU-Talk system has allowed potential users to view the interface design and to assess input devices and provide feedback on the how these could be altered or improved. Some of the patients who have had their communication observed in ICU and have viewed the interface designs for ICU-Talk, commented on how easy it is to learn to use and navigate. They have also commented favourably on colour schemes, font type and size. These users have also tested a range of input devices and enabled us to confirm that even with limited control of their upper limbs, they were able to use them effectively. Input devices assessed included mouse, drag-pad, trackball, as well as a range of single switches. Each patient had their preferred input device. The ability to accommodate individual patients preferences is an essential component in the development of ICU-Talk. Easy accessibility to adjust information and alter the performance of the system is needed to meet the ever-changing needs of the patient.

Conclusions and future work

Devising an AAC system specifically for patients in ICU is particularly challenging due to the patient's individual and complex needs and the limitations imposed by their condition and the ICU environment. Developing a database containing phrases that will be of most use to intubated patients who are temporarily unable to speak is central to the success of ICU-Talk.

Nursing staff from ICU were interviewed to identify those phrases used most frequently by patients to communicate. Patients who were attempting to communicate were then observed. Patients used a quarter of the phrases suggested by nursing staff. The majority of phrases communicated by patients were patient specific, reinforcing the need for an additional database of personalised phrases. This will be achieved with the co-operation of the patient's family and friends. A structured computerised interview, which will be used to assemble the relevant data and integrate it into the database.

The majority of phrases communicated by patients were under the topics of needs and feelings suggesting that patients are most concerned about their present circumstances and not the future.

The most common communication partner for the patient is the nurse. Within the ICU, nurses work 12 hour shifts and each nurse responsible for one patient. This allows a relationship to be built up between the patient and the nurse. Other members of staff tend to communicate via the nurse rather than directly with the patient. It is therefore essential to obtain a perception of the ICU-Talk system from the nursing staff. This will be achieved through questionnaires and interviews.

Friends and relatives have shared knowledge of the patient and how the patient tries to communicate to maintain their relationship and offer reassurance and support to the patient while in this unfamiliar and sometimes frightening environment. By communicating with friends and relatives the patient is able to keep in touch with "normal life" outside the ICU setting.

Patients use mouthing most frequently to communicate even though they are unable to vocalise. For many patients this is the most natural form of communication but it can be very difficult to lip read patients especially if they are orally intubated. The patient is unable to produce any sound therefore it can be difficult to attract attention, and when lip reading the communication partner has to be looking directly at the patient throughout the conversation. ICU-Talk will speak out the pre-stored phrases once they are selected making conversations more natural.

The interface design is ongoing and includes evaluation by potential users, nursing staff and relatives/partners of the patients. A full prototype system including the ability to include a patient specific database will be evaluated in ICU over a twelve month period and the results from this second phase of the project will be reported at future meetings.

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